

- which individual organisms meet the challenges posed by their biotic and abiotic environments.
- organ of Corti** (kor'-tē) The actual hearing organ of the vertebrate ear, located in the floor of the cochlear duct in the inner ear; contains the receptor cells (hair cells) of the ear.
- organogenesis** (ōr-gan'-ō-jen'-uh-sis) The process in which organ rudiments develop from the three germ layers after gastrulation.
- organ system** A group of organs that work together in performing vital body functions.
- origin of replication** Site where the replication of a DNA molecule begins, consisting of a specific sequence of nucleotides.
- orthologous genes** Homologous genes that are found in different species because of speciation.
- osculum** (os'-kyuh-lum) A large opening in a sponge that connects the spongocoel to the environment.
- osmoconformer** An animal that is isoosmotic with its environment.
- osmolarity** (oz'-mō-lār'-uh-tē) Solute concentration expressed as molarity.
- osmoregulation** Regulation of solute concentrations and water balance by a cell or organism.
- osmoregulator** An animal that controls its internal osmolarity independent of the external environment.
- osmosis** (oz-mō'-sis) The diffusion of free water across a selectively permeable membrane.
- osteichthyan** (os'-tē-ik'-thē-an) A member of a vertebrate clade with jaws and mostly bony skeletons.
- outer ear** One of the three main regions of the ear in reptiles (including birds) and mammals; made up of the auditory canal and, in many birds and mammals, the pinna.
- outgroup** A species or group of species from an evolutionary lineage that is known to have diverged before the lineage that contains the group of species being studied. An outgroup is selected so that its members are closely related to the group of species being studied, but not as closely related as any study-group members are to each other.
- oval window** In the vertebrate ear, a membrane-covered gap in the skull bone, through which sound waves pass from the middle ear to the inner ear.
- ovarian cycle** (ō-vār'-ē-un) The cyclic recurrence of the follicular phase, ovulation, and the luteal phase in the mammalian ovary, regulated by hormones.
- ovary** (ō'-vuh-rē) (1) In flowers, the portion of a carpel in which the egg-containing ovules develop. (2) In animals, the structure that produces female gametes and reproductive hormones.
- oviduct** (ō'-vuh-duct) A tube passing from the ovary to the vagina in invertebrates or to the uterus in vertebrates, where it is also called a fallopian tube.
- oviparous** (ō-vip'-uh-rus) Referring to a type of development in which young hatch from eggs laid outside the mother's body.
- ovoviviparous** (ō'-vō-vī-vip'-uh-rus) Referring to a type of development in which young hatch from eggs that are retained in the mother's uterus.
- ovulation** The release of an egg from an ovary. In humans, an ovarian follicle releases an egg during each uterine (menstrual) cycle.
- ovule** (ō'-vyūl) A structure that develops within the ovary of a seed plant and contains the female gametophyte.
- oxidation** The complete or partial loss of electrons from a substance involved in a redox reaction.
- oxidative phosphorylation** (fos'-fōr-uh-lā'-shun) The production of ATP using energy derived from the redox reactions of an electron transport chain; the third major stage of cellular respiration.
- oxidizing agent** The electron acceptor in a redox reaction.
- oxytocin** (ok'-si-tō'-sen) A hormone produced by the hypothalamus and released from the posterior pituitary. It induces contractions of the uterine muscles during labor and causes the mammary glands to eject milk during nursing.
- p53 gene** A tumor-suppressor gene that codes for a specific transcription factor that promotes the synthesis of proteins that inhibit the cell cycle.
- paedomorphosis** (pē'-duh-mōr'-fuh-sis) The retention in an adult organism of the juvenile features of its evolutionary ancestors.
- pain receptor** A sensory receptor that responds to noxious or painful stimuli; also called a nociceptor.
- paleoanthropology** The study of human origins and evolution.
- paleontology** (pā'-lē-un-tol'-ō-jē) The scientific study of fossils.
- pancreas** (pan'-krē-us) A gland with exocrine and endocrine tissues. The exocrine portion functions in digestion, secreting enzymes and an alkaline solution into the small intestine via a duct; the ductless endocrine portion functions in homeostasis, secreting the hormones insulin and glucagon into the blood.
- pancrustacean** A member of a diverse arthropod clade that includes lobsters, crabs, barnacles and other crustaceans, as well as insects and their six-legged terrestrial relatives.
- pandemic** A global epidemic.
- Pangaea** (pan-jē'-uh) The supercontinent that formed near the end of the Paleozoic era, when plate movements brought all the landmasses of Earth together.
- parabasalid** A protist, such as a trichomonad, with modified mitochondria.
- paracrine** Referring to a secreted molecule that acts on a neighboring cell.
- paralogous genes** Homologous genes that are found in the same genome as a result of gene duplication.
- paraphyletic** (pār'-uh-fī-let'-ik) Pertaining to a group of taxa that consists of a common ancestor and some, but not all, of its descendants.
- parareptile** A basal group of reptiles, consisting mostly of large, stocky quadrupedal herbivores. Parareptiles died out in the late Triassic period.
- parasite** (pār'-uh-sit) An organism that feeds on the cell contents, tissues, or body fluids of another species (the host) while in or on the host organism. Parasites harm but usually do not kill their host.
- parasitism** (pār'-uh-sit-izm) A +/- ecological interaction in which one organism, the parasite, benefits by feeding upon another organism, the host, which is harmed; some parasites live within the host (feeding on its tissues), while others feed on the host's external surface.
- parasympathetic division** A division of the autonomic nervous system; generally enhances body activities that gain and conserve energy, such as digestion and reduced heart rate.
- parathyroid gland** One of four small endocrine glands, embedded in the surface of the thyroid gland, that secrete parathyroid hormone.
- parathyroid hormone (PTH)** A hormone secreted by the parathyroid glands that raises blood calcium level by promoting calcium release from bone and calcium retention by the kidneys.
- parenchyma cell** (puh-ren'-ki-muh) A relatively unspecialized plant cell type that carries out most of the metabolism, synthesizes and stores organic products, and develops into a more differentiated cell type.
- parental type** An offspring with a phenotype that matches one of the true-breeding parental (P generation) phenotypes; also refers to the phenotype itself.
- Parkinson's disease** A progressive brain disease characterized by difficulty in initiating movements, slowness of movement, and rigidity.
- parthenogenesis** (par'-thuh-nō'-jen'-uh-sis) A form of asexual reproduction in which females produce offspring from unfertilized eggs.
- partial pressure** The pressure exerted by a particular gas in a mixture of gases (for instance, the pressure exerted by oxygen in air).
- passive immunity** Short-term immunity conferred by the transfer of antibodies, as occurs in the transfer of maternal antibodies to a fetus or nursing infant.
- passive transport** The diffusion of a substance across a biological membrane with no expenditure of energy.
- pathogen** An organism or virus that causes disease.
- pathogen-associated molecular pattern (PAMP)** A molecular sequence that is specific to a certain pathogen.
- pattern formation** The development of a multicellular organism's spatial organization, the arrangement of organs and tissues in their characteristic places in three-dimensional space.
- peat** Extensive deposits of partially decayed organic material often formed primarily from the wetland moss *Sphagnum*.
- pedigree** A diagram of a family tree with conventional symbols, showing the occurrence of heritable characters in parents and offspring over multiple generations.
- pelagic zone** The open-water component of aquatic biomes.
- penis** The copulatory structure of male mammals.
- PEP carboxylase** An enzyme that adds CO₂ to phosphoenolpyruvate (PEP) to form oxaloacetate in mesophyll cells of C₄ plants. It acts prior to photosynthesis.